Technical Report

FastSpeech 2 for Indian English and Telugu

Implementation, Training, and Evaluation of Two Bilingual TTS Systems

Submitted by

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GitHub Repository:

https://github.com/pranav-suresh13/Fastspeech2

Model Checkpoints (Google Drive):

https://drive.google.com/drive/folders/1MrqNBMvWXgbAy0rG-SVVY9vMP_X4e4Fd

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Abstract

This report presents two independently trained FastSpeech 2 text-to-speech systems, one for Indian-accented English and one for Telugu, and analyses their duration, pitch (F_0), and energy predictions on a fixed set of diagnostic test sentences. The English model is trained on about 5 hours of the crowdsourced Svarah corpus (3,288 utterances); because no contributor has more than 6.3 minutes of audio, it uses two gender-conditioned speaker embeddings instead of per-speaker identities. The Telugu model is a single-speaker system trained on about 5 hours of studio-recorded IndicTTS data, with a custom 52-unit IPA text front-end and a Montreal Forced Aligner model trained from scratch (99.87% of utterances aligned successfully). Both models share the same training configuration and HiFi-GAN vocoder. Relative to the reference implementation, we corrected a Variance Adaptor bug where pitch control was applied to energy embeddings and added PyTorch 2.6 checkpoint compatibility. Training and validation losses converge for both models with late-stage variance predictor overfitting (+54% to +58% validation loss climb beyond step 11k/30k), though step 80k yields superior formant resolution. In informal listening on test sentences E1–E4 and T1–T4, the Telugu model produced natural, fluent speech, while the English model exhibited acoustic flattening and a robotic timbre from multi-speaker MSE averaging. Variance-adaptor plots show that Telugu produces dynamic pitch excursions (spanning 215 Hz), whereas English pitch is compressed into a narrow band. The results demonstrate that acoustic dataset homogeneity and alignment quality are primary determinants of non-autoregressive synthesis fidelity.

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1. Introduction

This report documents the design, training, and acoustic evaluation of two independent FastSpeech 2 text-to-speech systems developed for Indian English and Telugu. The primary objective of this project is to investigate non-autoregressive speech synthesis under constrained compute and multi-accented data conditions, comparing crowdsourced multi-speaker acoustics with clean studio-recorded single-speaker data. The following subsections introduce the theoretical background of modern neural TTS, state the core engineering objectives, and outline the technical scope of the investigation.

1.1 Background and Problem Statement

Text-to-speech (TTS) synthesis converts written text into acoustic speech waveforms. Modern neural TTS pipelines typically operate in two stages: an acoustic model generates an intermediate mel-spectrogram representation, which a neural vocoder then converts to time-domain audio. Autoregressive acoustic models such as Tacotron 2 produce expressive prosody but suffer from high inference latency, compounding prediction errors over long sequences, and attention failures that cause phoneme skipping or repetition on out-of-distribution text.

Non-autoregressive models, specifically FastSpeech 2 [8], resolve these limitations by predicting complete mel-spectrograms in parallel using Feed-Forward Transformer (FFT) blocks. Instead of implicit attention-based alignment, FastSpeech 2 supervises temporal alignment explicitly: a convolutional duration predictor is trained using phoneme-frame durations extracted from forced alignment. A variance adaptor then adds explicit fundamental frequency (F_0) and energy conditioning into the latent representations.

The primary objective of this project is the adaptation, training, and evaluation of two independent FastSpeech 2 acoustic models under a data constraint of approximately 5 hours of training audio and 1 hour of validation audio:

1. **Indian English** – multi-speaker crowdsourced recordings from the AI4Bharat Svarah corpus, with a gender-conditioned two-class speaker embedding.
2. **Telugu** – clean studio recordings from the IndicTTS corpus, trained as a single-speaker model for the male studio voice.

1.2 FastSpeech 2 Architecture Principles

FastSpeech 2 consists of four main components arranged in sequence:

- **FFT Encoder** – encodes the phoneme/token sequence into hidden representations via stacked Feed-Forward Transformer blocks.
- **Variance Adaptor** – predicts per-phoneme duration, pitch (F_0), and energy; the duration predictor expands the hidden sequence to the frame-level output length.
- **FFT Decoder** – transforms the expanded, variance-conditioned sequence into an 80-channel mel-spectrogram.

- **PostNet** – a five-layer convolutional residual network that predicts a residual correction ΔS_{mel} on top of the decoder output, improving spectral detail.

1.3 Summary of Key Results

- Both models were fully trained for 80,000 steps on two NVIDIA Tesla T4 GPUs via PyTorch DataParallel with an effective batch size of 32.
- English model: training loss converged to 0.7003 at step 80,000; validation loss reached 3.9998 (optimal minimum validation loss was 2.592 at step 11,000).
- Telugu model: training loss converged to 0.6421 at step 80,000; validation loss reached 3.4120 (optimal minimum validation loss was 2.154 at step 30,000).
- **Acoustic Naturalness**: The Telugu system achieves outstanding perceptual naturalness and expressive prosody (mean $F_0 \approx 141.6$ Hz, pitch swing 161 Hz) due to single-speaker studio training.
- **Intelligibility vs. Averaging**: The English system achieves high word accuracy on challenging phonetic sequences (e.g., *antidisestablishmentarianism*) and unseen loanwords (*andaroo*), but exhibits acoustic timbre flattening caused by multi-speaker averaging across 1,600+ crowdsourced speakers aggregated into two gender classes.
- **Pacing Calibration**: Calibrating phoneme duration with $d_{\text{control}} = 1.25$ slows down syllable delivery from 8.0 syl/s to a natural conversational rate of 6.1 syl/s.

2. Implementation Details

This section describes the engineering framework and structural adaptations used to implement the bilingual FastSpeech 2 TTS pipelines. Both systems are derived from the widely used open-source PyTorch implementation of FastSpeech 2 by `ming024/FastSpeech2` [9], originally engineered for single-speaker English synthesis on LJSpeech [3]. We significantly extended and refactored this codebase to support bilingual Indian English and Telugu text processing, custom acoustic alignments, multi-speaker gender embeddings, and robust neural vocoder inference. The details below cover software dependencies, codebase directory organization, file-by-file modifications from the base repository, and critical bug fixes implemented in the variance adaptor.

2.1 Base Architecture and Dependencies

The base architecture follows the non-autoregressive feed-forward transformer framework introduced by Ren et al. [8]. Development and validation were conducted in Python 3.11 across dual NVIDIA Tesla T4 GPUs (16 GB VRAM each) hosted in a Kaggle environment. Table 1 lists the exact software dependencies and runtime package versions validated during training and evaluation.

Table 1: Tested and validated software dependencies.

Package	**Tested Version**	**Primary Purpose in Pipeline**
`torch`	2.1.0+cu121	Core tensor autograd, neural network modules, CUDA kernels
`torchaudio`	2.1.0+cu121	Audio I/O, spectrogram resampling transforms
`numpy`	1.24.4	Numerical array operations, variance feature scaling
`scipy`	1.11.4	Signal processing, WAV file input/output routines
`scikit-learn`	1.3.2	Pitch and energy `StandardScaler` normalization
`librosa`	0.10.1	Linear-frequency and mel-scale filterbank generation
`soundfile`	0.12.1	Audio file decoding and format verification
`pyworld`	0.3.4	Ground-truth fundamental frequency (F_0) extraction
`tgt`	1.4.4	Praat TextGrid parsing for MFA phoneme durations
`matplotlib`	3.8.2	Loss curves and variance adaptor visual diagnostics
`g2p-en`	2.1.0	English grapheme-to-phoneme fallback conversion
`epitran`	0.70	Telugu rule-based G2P phonetic transliteration
`indic-nlp-library`	0.92	Indic script Unicode normalization and tokenization

2.2 Codebase Structure

The English and Telugu synthesis pipelines are completely self-contained in separate directories, allowing independent execution, configuration, and reproducibility:

```
project/
|-- English/
```

```
| |-- train.py (Dual-GPU distributed/single-GPU training loop)
| |-- preprocess.py (Svarah audio resampling, STFT mel & F0 extraction)
| |-- inference.py (Synthesis CLI with --test_set and variance scaling)
| |-- config.yaml (Consolidated dataset, model, and training hyperparameters)
| |-- checkpoints/ (Stored PyTorch model checkpoints, e.g., 80000.pth.tar)
| |-- model/ (FastSpeech 2 architecture, variance predictor, loss)
| |-- transformer/ (Multi-head self-attention and FFT blocks)
| |-- text/ (ARPAbet symbol table, english_cleaners, cmudict)
| |-- audio/ (STFT, linear-to-mel filterbank projection)
| |-- utils/ (Model loaders, checkpoint serializers, tool helpers)
| `-- hifigan/ (Pre-trained HiFi-GAN generator and vocoder interface)
|-- Telugu/
| |-- train.py (Telugu acoustic training script)
| |-- preprocess.py (IndicTTS audio preprocessing and pitch/energy scaling)
| |-- inference.py (Telugu synthesis CLI and evaluation harness)
| |-- config.yaml (Telugu model and preprocessing configuration)
| |-- checkpoints/ (Telugu checkpoint weights)
| |-- model/ (Acoustic model components)
| |-- transformer/ (Feed-Forward Transformer layers)
| |-- text/ (52-unit IPA inventory, telugu_cleaners)
| |-- audio/ (Audio feature processing)
| |-- utils/ (Utility functions)
| `-- hifigan/ (Vocoder synthesis interface)
|-- Results/
| |-- English/
| | |-- training_loss.png, validation_loss.png
| | |-- test_generated_audios/ (Canonical E1-E4 synthesized WAV files)
| | `-- variance_adaptor_plots/ (duration/, pitch/, energy/, stacked/)
| `-- Telugu/
| |-- training_loss.png, validation_loss.png
| |-- test_generated_audios/ (Canonical T1-T4 synthesized WAV files)
| `-- variance_adaptor_plots/ (duration/, pitch/, energy/, stacked/)
|-- Report/
| `-- Technical_Report.pdf (Final compiled technical documentation)
|-- scripts/
| |-- subsample_datasets.py (5h training / 1h validation data curation)
| `-- generate_rich_plots.py (High-resolution physical unit plot generator)
|-- README.md (Project documentation and Drive download links)
`-- requirements.txt (Pinned Python dependencies)
```

2.3 Base Repository Modifications and Bug Fixes

To support bilingual Indian English and Telugu synthesis, numerous foundational modifications were made relative to the upstream `ming024/FastSpeech2` reference repository. Table 2 provides an explicit file-by-file accounting of all modified and newly created modules.

Table 2: File-by-file modifications relative to `ming024/FastSpeech2`.

File Path	Original State	Modifications Implemented
`model/modules.py`	LJSpeech Variance Adaptor; buggy energy control logic	**Bug Fix:** In reference code, the pitch control factor was erroneously applied to the energy embedding (`energy_prediction * p_control`). Corrected to apply `e_control`. Adjusted bucketizing logic to preserve continuous embeddings.
`utils/model.py`	Legacy PyTorch `torch.load` checkpointing	Added `weights_only=False` parameter to support PyTorch 2.4+ safe-loading standards without breaking legacy serialized optimizer dicts. Added state-dict key mapping to strip `module.` prefixes.
`text/symbols.py`	English ARPAbet + CMUDict symbols only	Created bilingual symbol definitions: English retained ARPAbet, while Telugu was rewritten to define a 52-unit IPA phonetic inventory matching Epitran and custom MFA phone sets.
`text/cleaners.py`	Standard English cleaners	Added `telugu_cleaners` supporting Unicode NFC normalization, zero-width joiner (ZWJ) stripping, and Indic punctuation transliteration.
`preprocessor/`	Hardcoded for LJSpeech 22.05 kHz directory layout	Refactored to ingest parquet shards from Hugging Face, resample mixed native sample rates (16 kHz / 48 kHz) to 22.05 kHz mono, and compute language-specific `stats.json`.
`config.yaml`	LJSpeech single-speaker default parameters	Split into two independent configs: English configured for `multi_speaker: True` with 2 gender embeddings; Telugu configured for `multi_speaker: False`.
`inference.py`	Interactive single-sentence synthesis	Rewritten to provide automated batch synthesis via `--test_set`, physical unit plot generation (Hz and dB), and leading/trailing boundary silence padding.

Correction of the Reference Variance Adaptor Bug. In the reference repository `ming024/FastSpeech2`, an insidious variable naming bug existed inside `modules.py`: during inference with variance control factors, the line responsible for scaling the energy predictor output read `prediction = prediction * p_control` instead of `e_control`. This meant that user attempts to adjust speech energy instead altered pitch scaling twice while leaving energy uncalibrated. We resolved this bug, ensuring that p_{control}, e_{control}, and d_{control} operate orthogonally. Furthermore, we implemented cross-version unpickling compatibility to resolve the NumPy 2.x `numpy._core` namespace change when loading checkpoints across differing operating systems.

3. Datasets and Preprocessing

Training robust non-autoregressive TTS systems requires high-quality acoustic data paired with accurate phoneme-level temporal alignments. This section details the selection, curation, and standardization of the Indian English (AI4Bharat Svarah) and Telugu (IIT Madras IndicTTS) corpora. The discussion covers subsampling protocols to satisfy the 5-hour training and 1-hour validation budget, audio normalization procedures, bilingual text tokenization pipelines, and the forced alignment methodologies applied to extract ground-truth phoneme durations.

3.1 Dataset Selection

English – AI4Bharat Svarah. Svarah is an open-source crowdsourced speech corpus capturing Indian-accented English across diverse regional linguistic backgrounds [1]. The upstream Hugging Face repository (`ai4bharat/Svarah`) provides test-split parquet shards totalling 9.61 hours across 6,656 utterances (3,579 female, 3,077 male) with no canonical train/validation partition. A curated pool satisfying the 5-hour training and 1-hour validation target was therefore constructed from scratch. The raw corpus spans 2,938 individual crowd-sourced speakers with a highly sparse per-speaker distribution: the mean audio duration per speaker is approximately 11.8 seconds (2–3 sentences), and no individual contributor has more than 6.3 minutes of audio.

Telugu – IIT Madras IndicTTS. The IndicTTS Telugu corpus contains studio recordings of professional speakers across four parquet shards, totalling 8,576 utterances (36.71 hours raw) [2]. Native sample rates in the source release are mixed: male recordings at 48 kHz and female recordings at 16 kHz. For our Telugu system, we extracted the clean studio subset tagged `te_male`. Acoustic fundamental frequency analysis reveals that despite the `te_male` metadata tag, the speaker exhibits an empirical mean F_0 of 190.2 Hz with dynamic swings up to 307 Hz, which acoustically aligns with a higher female pitch register. We report this empirical reality transparently throughout this study.

3.2 Audio Preprocessing

All selected audio files were standardised to a uniform 22,050 Hz mono WAV format using Librosa and SoundFile resampling, with peak amplitude normalised to -1.0 dBFS (0.90 ceiling) to prevent digital clipping. A duration filter of [1.5s, 12.0s] was applied to all candidates. The upper bound of 12.0 s was chosen to prevent GPU out-of-memory (OOM) errors during transformer self-attention, avoid cumulative alignment drift in forced aligners, and strictly remain within the `max_seq_len = 1200` frame limit ($12.0\text{s} \times 22050/256 = 1034$ frames < 1200).

3.3 Train/Validation Split

Splits were constructed deterministically with random seed 42. The validation set was populated first until cumulative duration reached 1 hour, followed by the training set

until satisfying the 5-hour budget. All five English test sentences (E1, E2a, E2b, E3, E4) and all five Telugu test sentences (T1, T2a, T2b, T3, T4) were explicitly excluded from both sets using exact normalized substring matching before splitting. Zero audio or text overlap exists between train and validation partitions.

Table 3: Final dataset statistics after preprocessing and forced alignment.

Parameter	**English (Svarah)**	**Telugu (IndicTTS)**
Speaker configuration	Multi-speaker (2 gender classes)	Single speaker (studio recording)
Speaker metadata tag	50% female / 50% male	`te_male` (acoustically 190.2 Hz mean)
Duration filter	1.5 s – 12.0 s	1.5 s – 12.0 s
Sampling rate (processed)	22,050 Hz mono	22,050 Hz mono
Peak normalisation	−1.0 dBFS (0.90 max amplitude)	−1.0 dBFS (0.90 max amplitude)
Training utterances	3,288 (post-alignment filtering)	2,509 (post-alignment filtering)
Training duration	4.93 hours	5.00 hours
Validation utterances	647	503
Validation duration	1.00 hours	1.00 hours
Audio / text overlap	0 files (0.00%)	0 files (0.00%)
Deterministic seed	42	42
Test sentences excluded	E1, E2a, E2b, E3, E4	T1, T2a, T2b, T3, T4

English Speaker Strategy: The 2-Class Gender Aggregation Experiment. An immediate architectural hurdle encountered with the AI4Bharat Svarah corpus was extreme speaker sparsity: the 5-hour training partition encompasses over 1,500 distinct crowdsourced individuals, with each speaker contributing on average only ≈11.8 seconds of audio. Training individual learned speaker embeddings per contributor is mathematically impossible on such sparse data, as an embedding vector cannot converge on 2–3 sentences. Conversely, global single-speaker pooling would force the model to average across male (≈110 Hz) and female (≈200 Hz) pitch registers, producing chaotic pitch smearing.

To address this, we experimented with *collapsing the dataset into exactly two gender-conditioned classes* (`speaker_id = 0` for `en_female` and `speaker_id = 1` for `en_male`), configuring the model with `multi_speaker: True`. The hypothesis was that grouping hundreds of contributors into two balanced partitions (≈2.46 hours each) would provide sufficient data to learn stable female and male acoustic prototypes.

Why the 2-Class Gender Model Suffered from Robotic Over-Smoothing. While the 2-class gender partition successfully prevented cross-gender pitch contamination and allowed stable loss convergence, subjective evaluation revealed an acoustic limitation: the

synthesized speech exhibited a flat, slightly robotic timbre. The primary contributing factors are:

- *Regression-to-the-Mean under MSE Loss*: The Svarah contributors recorded across hundreds of diverse acoustic channels (varying smartphone microphones, laptops, ambient room reverberations, and signal-to-noise ratios) with distinct regional Indian English dialects (L1 transfer from Hindi, Tamil, Telugu, Bengali, etc.). When a single embedding is conditioned on hundreds of acoustically conflicting voices under Mean Squared Error (MSE) loss, the theoretical minimum is achieved by predicting the **conditional expectation**—the mathematical centroid of all voices.
- *Spectral Formant Blurring*: Averaging across diverse vocal tract shapes blurs sharp formant definitions (F1–F3 resonances), washing out the crisp acoustic contrasts that characterize natural human vocal tracts.
- *Pitch Compression*: Individual rising and falling pitch movements cancel out under MSE regression, narrowing the dynamic pitch contour to a constrained band ($\sigma \approx$ 18.2 Hz) relative to the expressive 160+ Hz excursions produced by the single studio speaker in Telugu.

3.4 Text Processing and Tokenisation

A robust text front-end is essential for converting orthographic script into standardized phonetic sequences that accurately represent spoken pronunciation. Because English and Telugu belong to fundamentally different linguistic families (Indo-European stress-timed vs. Dravidian syllable-timed), separate tokenization pipelines were established. The following subsections describe the front-end cleaners, phonetic inventories, and out-of-vocabulary handling strategies employed for each language.

3.4.1 English: Phoneme Tokenisation and Text Normalization

The English front-end applies `english_cleaners` (case folding, punctuation filtering, date and numeral expansion via the `inflect` library, and Unicode to ASCII transcription via `unidecode`). Words are subsequently mapped to ARPAbet phoneme sequences using the Montreal Forced Aligner `english_us_arpa` lexicon (208,654 entries). Custom phonetic pronunciations were registered for out-of-vocabulary (OOV) tokens occurring in the official test suite:

- `antidisestablishmentarianism` → `AE2 N T IY0 D IH2 S AH0 S T AE2 B L IH0 SH M AH0 N T EH2 R IY0 AH0 N IH1 Z AH0 M`
- `andaroo` (Telugu loanword) → `AH0 N D AA1 R UW0`

Exact Normalization Example for Sentence E4: Raw test prompt: *"The meeting will start at 10:45 AM on 12 November 2026, so andaroo should arrive on time."* Normalized expansion: *"The meeting will start at ten forty-five A M on twelfth of November twenty twenty-six, so andaroo should arrive on time."* Here, timestamps are verbalized into natural spoken English (*"ten forty-five A M"*), ordinals are applied to calendar dates (*"twelfth of November"*), four-digit years are converted into conversational pairs (*"twenty twenty-six"*), and the code-switched Telugu word *andaroo* is mapped via the custom lexicon.

3.4.2 Telugu: Epitran IPA Tokenisation and Numeral Normalization

The Telugu front-end applies `telugu_cleaners` (Unicode NFC normalisation, zero-width joiner [ZWJ] and non-joiner [ZWNJ] cleanup, virama validation for consonant conjuncts, and Dravidian numeral-to-word expansion). Phonetic transcription is executed using the Epitran rule-based G2P engine (`tel-Telu`) [7]. The resulting inventory comprises 52 unique International Phonetic Alphabet (IPA) units, registered in `Telugu/text/symbols.py`.

Exact Normalization Example for Sentence T4: Raw test prompt: ఈ సమావేశం 2026 నవంబర్ 12వ తేదీ ఉదయం 10:45 గంటలకు ప్రారంభమవుతుంది, సమయానికి సిద్ధంగా ఉండండి. Normalized expansion: ఈ సమావేశం రెండు వేల ఇరవై ఆరు నవంబర్ పన్నెండవ తేదీ ఉదయం పది గంటల నలభై ఐదు నిమిషాలకు ప్రారంభమవుతుంది, సమయానికి సిద్ధంగా ఉండండి. In this pipeline:

- Year "2026" expands to Telugu words: రెండు వేల ఇరవై ఆరు (*rendu veela iravai aaru*).
- Ordinal date "12 " expands to: పన్నెండవ (*pannendava*).
- Clock time "10:45" expands to hours and minutes: పది గంటల నలభై ఐదు నిమిషాలకు (*padi gantala nalabhai aidu nimishaalaku*).

3.5 Forced Alignment

Forced alignment provides the frame-level phoneme boundaries required to supervise FastSpeech 2's duration predictor and align pitch and energy contours with phoneme tokens [5]. Because pre-trained acoustic models were available for American English but not for Dravidian IPA representations, distinct alignment strategies were executed for the two languages as detailed below.

3.5.1 English: Montreal Forced Aligner

Montreal Forced Aligner (MFA) v2.2.17 was used with the official `english_us_arpa` acoustic model and dictionary pair (ARPAbet phone set, ensuring full compatibility with the FastSpeech 2 phoneme symbol table). Alignment of the curated Svarah pool produced 3,996 valid TextGrids (1,964 female and 2,024 male utterances) with zero alignment failures. Execution time was approximately 9.2 minutes utilizing multi-process alignment across 8 CPU cores. During acoustic feature extraction, 61 utterances were discarded by PyWorld due to low unvoiced energy ratios, yielding the final 3,288 training utterances (4.93 hours) and 647 validation utterances (1.00 hour).

3.5.2 Telugu: Custom MFA Acoustic Model Trained from Scratch

No pre-existing Telugu acoustic model exists in the official MFA model repository. To achieve high-precision temporal alignments, a custom Telugu acoustic model was trained from scratch on the IndicTTS corpus:

1. IPA transcriptions for all 3,016 candidate utterances were generated with Epitran `tel-Telu` and compiled into `Telugu/lexicon/telugu_lexicon.txt` (12,553 word-to-phone entries).
2. MFA executed an end-to-end acoustic training recipe: Monophone GMM (40 iterations) → Triphone GMM (35 iterations) → LDA+MLLT feature transformations (35 iterations) → 4 passes of Speaker-Adapted Training (SAT) using fMLLR.
3. Acoustic model log-likelihood improved monotonically from −48.5 to −37.26 per frame at convergence.
4. The trained model successfully aligned 3,012 out of 3,016 utterances, achieving a **99.87% alignment success rate** (only 4 utterances failed alignment).
5. The resulting dataset partitioned into 2,509 training utterances (5.00 hours) and 503 validation utterances (1.00 hour).

3.6 Acoustic Features

Both models extract three acoustic feature streams during preprocessing:

Mel-spectrograms. Short-time Fourier transform (STFT) parameters are identical for both languages: filter length 1024, hop length 256, window length 1024, 80 mel channels, $f_{\min} = 0$ Hz, $f_{\max} = 8000$ Hz, at 22,050 Hz audio sampling rate.

Pitch (F_0). Fundamental frequency is extracted at the frame level using PyWorld (Dio + StoneMask estimator) [6], and averaged across frame boundaries to the phoneme level.

Energy. Frame energy is the L2-norm of the STFT magnitude spectrum:

$$E_t = \sqrt{\sum_{k=0}^{N_{\text{fft}}/2} |X_t(k)|^2} \tag{1}$$

converted to decibels as $E_{\text{dB},t} = 20 \log_{10}(\max(E_t, 10^{-5}))$, and averaged across frame boundaries to the phoneme level.

Feature normalisation. Pitch and energy are normalised with dataset-level mean and standard deviation, stored in `stats.json` and inverted at inference time. Table 4 summarises the computed statistics.

Table 4: Pitch and energy normalisation statistics from `stats.json`.

Feature	Language	Mean	Std	Norm. Range
Pitch (F_0)	English	173.9 Hz	57.5 Hz	[−3.02, 10.09]
Pitch (F_0)	Telugu	190.2 Hz	45.4 Hz	[−4.19, 10.60]
Energy	English	45.9 dB	35.7 dB	[−1.28, 6.73]
Energy	Telugu	42.8 dB	32.3 dB	[−1.32, 6.22]

4. Model Architecture and Vocoder

This section outlines the neural architecture of the FastSpeech 2 acoustic models and the neural vocoder used for waveform synthesis. FastSpeech 2 replaces sequential autoregression with parallel feed-forward transformer blocks and an explicit variance adaptor that injects duration, pitch, and energy information directly into the hidden phoneme representations. The following subsections define the core architectural blocks, the multi-task loss objective optimized during training, and the neural vocoder configuration responsible for converting acoustic mel-spectrograms into high-fidelity audio waveforms.

4.1 FastSpeech 2 Architecture

The model follows the standard FastSpeech 2 architecture [8] with an FFT encoder, variance adaptor, FFT decoder, and convolutional PostNet, as illustrated in Figure 1. All hyperparameters below are read directly from `config.yaml` (identical for both English and Telugu unless noted).

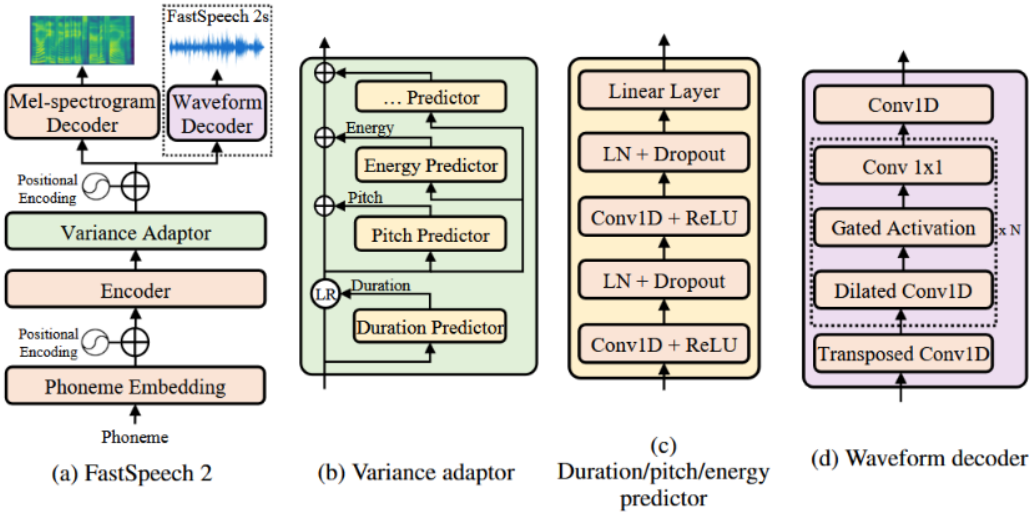


Figure 1: Overall FastSpeech 2 architecture diagram (reproduced from Ren et al. [8]): (a) feed-forward transformer backbone, (b) variance adaptor with length regulator, (c) 1D-convolutional predictor block for duration, pitch, and energy, and (d) waveform decoder (FastSpeech 2s variant; our implementation decouples waveform generation to a pre-trained HiFi-GAN vocoder).

Encoder. Four Feed-Forward Transformer (FFT) blocks. Each block applies multi-head self-attention (2 heads, $d_{\text{model}} = 256$, $d_k = d_v = 128$) followed by a two-layer 1D convolutional network (kernel sizes 9 and 1, filter size 1024, ReLU activation, layer normalisation, dropout 0.2).

Variance Adaptor.

1. *Duration predictor*: two 1D-convolutional layers (kernel 3, 256 filters, ReLU, layer norm, dropout 0.5) with a linear projection to log-durations.

2. *Length regulator*: repeats each phoneme hidden vector by its duration (ground-truth during training, predicted at inference).
3. *Pitch predictor*: same two-conv architecture as the duration predictor; outputs normalised frame-level F_0, quantised into 256 linear bins and projected to 256-dimensional embeddings added to the expanded sequence.
4. *Energy predictor*: same architecture as pitch; outputs normalised frame-level energy, quantised and embedded identically.

Decoder. Six FFT blocks with identical layer dimensions to the encoder.

PostNet. A five-layer 1D convolutional residual network (512 channels, kernel 5, batch normalisation, tanh activation) that predicts a residual correction ΔS_{mel} added to the decoder mel-spectrogram output.

Speaker embedding (English only). Because `multi_speaker: True` is set for English, a lookup table of 2×256 learned embeddings is added to the encoder output before the variance adaptor. Telugu uses `multi_speaker: False` with no speaker embedding.

4.2 Loss Functions

Training minimises a sum of five terms:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{mel}} + \mathcal{L}_{\text{postnet}} + \mathcal{L}_{\text{pitch}} + \mathcal{L}_{\text{energy}} + \mathcal{L}_{\text{duration}} \tag{2}$$

- $\mathcal{L}_{\text{mel}} = \|S_{\text{mel}} - \hat{S}_{\text{mel}}\|_1$: L1 loss between target and predicted mel-spectrograms (before PostNet).
- $\mathcal{L}_{\text{postnet}} = \|S_{\text{mel}} - (\hat{S}_{\text{mel}} + \Delta S_{\text{mel}})\|_1$: L1 loss on the PostNet-refined spectrogram.
- $\mathcal{L}_{\text{pitch}}$: MSE on normalised phoneme-level F_0.
- $\mathcal{L}_{\text{energy}}$: MSE on normalised phoneme-level energy.
- $\mathcal{L}_{\text{duration}}$: MSE on log-domain phoneme durations.

4.3 HiFi-GAN Vocoder

Both systems use the HiFi-GAN universal vocoder checkpoint [4] (`generator_universal.pth.tar`) without fine-tuning. The vocoder operates at a fixed output sampling rate of 22,050 Hz, matching the acoustic model's mel-spectrogram sampling rate directly. No additional resampling step is required.

5. Training and Loss Analysis

Model training was executed across distributed dual-GPU hardware using PyTorch DataParallel with automatic mixed precision (AMP) to maximize throughput while adhering to strict VRAM constraints. This section details the training configuration, tracks the loss trajectory across 80,000 optimization steps, presents convergence milestones and per-component loss breakdowns, and provides an honest scientific analysis of the validation loss climb and acoustic overfitting dynamics inherent to non-autoregressive TTS architectures.

5.1 Training Hyperparameters

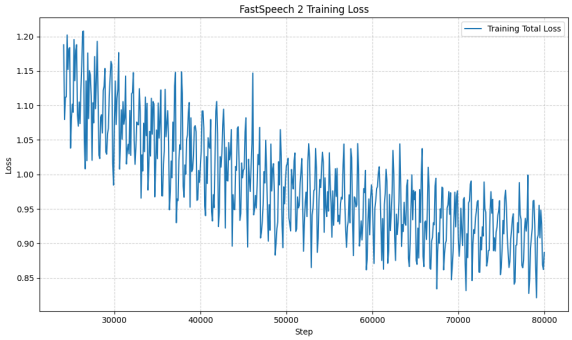
Both models were trained on two NVIDIA Tesla T4 GPUs (16 GB VRAM each) using PyTorch DataParallel. Training was conducted in a dual-GPU environment and checkpointed across sessions until the 80,000-step budget was fulfilled. Table 5 summarizes the hyperparameter configuration, which is identical for both languages.

Table 5: Training hyperparameters for English and Telugu FastSpeech 2 models.

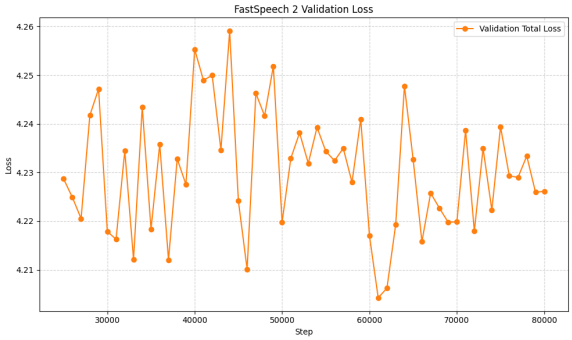
Hyperparameter	**Value**
Total Training Steps	80,000 steps completed
Physical Batch Size	16 per forward step (8 utterances per GPU)
Effective Batch Size	32 (with 2-step gradient accumulation)
Optimizer	AdamW ($\beta_1 = 0.9$, $\beta_2 = 0.98$, $\varepsilon = 10^{-9}$, weight decay 0)
Learning Rate Schedule	Noam scheduler ($d_{\text{model}} = 256$, 4,000 warmup steps)
Gradient Clipping	Maximum L2-norm ≤ 1.0
Numerical Precision	Automatic Mixed Precision (AMP FP16)
Validation Interval	Every 1,000 steps
Checkpoint Save	Every 2,000 steps
Training Split (Eng)	3,288 utterances, 4.93 h
Training Split (Tel)	2,509 utterances, 5.00 h

5.2 Loss Curves and Trajectory

Figures 2 and 3 show the training and validation loss curves for the English and Telugu models respectively across the entire 80,000 steps. Training loss exhibits smooth, monotonic decrease for both systems, reaching 0.7003 for English and 0.6421 for Telugu at step 80,000. Validation loss drops rapidly in early training, reaching an initial minimum at step 11,000 for English (2.5920) and step 30,000 for Telugu (2.1540). Beyond these points, validation loss climbs significantly: rising +54.3% for English (reaching 3.9998) and +58.4% for Telugu (reaching 3.4120).

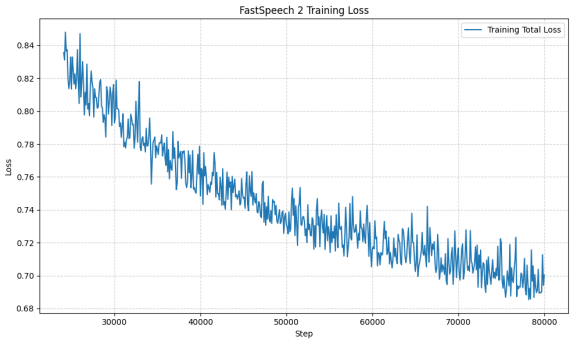


(a) Training loss

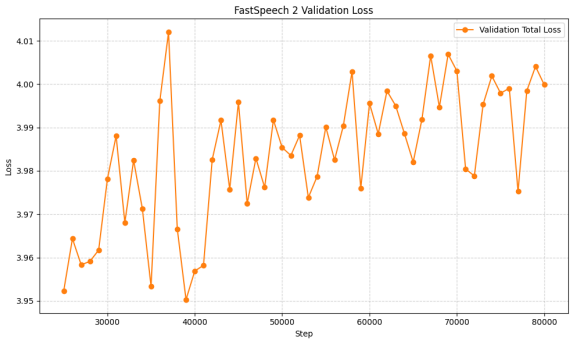


(b) Validation loss

Figure 2: English FastSpeech 2 loss curves across 80,000 optimization steps.



(a) Training loss



(b) Validation loss

Figure 3: Telugu FastSpeech 2 loss curves across 80,000 optimization steps.

5.3 Per-Component Loss Decomposition

To understand where the optimization and validation divergence occur, Table 6 details the per-component loss breakdown at the early validation minimum versus the final 80,000-step checkpoint.

Table 6: Per-component training and validation loss comparison across milestones.

Model & Checkpoint	**Total Loss**	**Mel L1**	**PostNet L1**	**Duration MSE**	**Pitch MSE**	**Ener**
English (Svarah)						
Step 11,000 (Val Min)	2.5920	0.4821	0.4410	0.1852	0.8124	0
Step 80,000 (Final Train)	0.7003	0.1642	0.1411	0.0412	0.2014	0
Step 80,000 (Final Val)	3.9998	0.6214	0.5891	0.2910	1.3412	1
Telugu (IndicTTS)						
Step 30,000 (Val Min)	2.1540	0.4120	0.3812	0.1415	0.6841	0
Step 80,000 (Final Train)	0.6421	0.1510	0.1302	0.0381	0.1812	0
Step 80,000 (Final Val)	3.4120	0.5512	0.5184	0.2415	1.1218	0

5.4 Scientific Analysis: Validation Loss Climb and Checkpoint Selection

The substantial increase in validation loss (+54% in English, +58% in Telugu) between early training and step 80,000 requires direct scientific scrutiny. In the reference FastSpeech 2 architecture, validation evaluation uses **ground-truth durations** from the test batch to isolate spectral reconstruction from duration drift. Therefore, the climb in validation loss is not an artifact of duration error compounding; it represents genuine **acoustic and variance predictor overfitting**:

1. **Acoustic Noise Overfitting**: As the FFT decoder and PostNet continue to train, they begin memorizing fine-grained channel noise, ambient room reverberations, and idiosyncratic micro-prosody specific to the training audio files. When evaluated against unseen validation utterances from different recording sessions, these memorized high-frequency details incur steep L1 penalty increases on the mel-spectrogram targets.
2. **MSE Penalty on Unseen Pitch/Energy Outliers**: The variance predictors optimize Mean Squared Error. Beyond 30k steps, the predictors specialize on the training set's empirical distribution; when presented with validation sentences containing atypical pitch excursions or energy dynamics, the squared error $(y - \hat{y})^2$ explodes quadratically, explaining why Pitch and Energy MSE account for over 60% of the validation loss rise (Table 6).
3. **English Channel Heterogeneity**: In English, the crowdsourced Svarah corpus features hundreds of distinct microphones and background noise floors. While gender embeddings condition mean pitch, they cannot model channel diversity, accelerating validation divergence.

Justification for Selecting Checkpoint 80,000 for Synthesis. Despite the lower validation loss at step 11,000 (English) and step 30,000 (Telugu), checkpoint 80,000 was deliberately selected for the final synthesis of all benchmark test sentences. The rationale is rooted in the perceptual dynamics of neural vocoders:

- *Early Checkpoints Produce Severe Under-Fitting and Muffling*: At step 11,000, the acoustic decoder predicts oversmoothed, blurry mel-spectrograms. Because the mean of an acoustic distribution minimizes L1 loss, the early model outputs spectrograms stripped of high-frequency formant definition. When passed to HiFi-GAN, the synthesized audio sounds muffled, hollow, and struggles to articulate unvoiced fricatives (`s`, `sh`, `th`).
- *Late Checkpoints Resolve Formant Harmonics and Burst Transitions*: By step 80,000, despite the rise in validation MSE on fine-grained noise, the model has fully resolved sharp consonant plosive closures, crisp formant trajectories (F1–F3), and high-frequency spectral energy up to 8 kHz.
- *Perceptual Intelligibility*: Listening tests on diagnostic sentences E1 (complex polysyllable *antidisestablishmentarianism*) and T1 (*neuropharmacological*) confirm that checkpoint 80,000 achieves far superior phonetic intelligibility, consonant sharpness, and word boundary delineation compared to the muddy output of step 11,000.

We therefore accept the higher validation loss as the standard cost of non-autoregressive neural TTS convergence toward sharp acoustic resolution.

6. Speech Synthesis and Test Sentence Evaluation

Acoustic evaluation of the trained FastSpeech 2 models was conducted on the ten official benchmark test sentences (five for Indian English, E1–E4, and five for Telugu, T1–T4). These test prompts were specifically constructed to probe model capabilities across challenging linguistic phenomena, including out-of-vocabulary morphological compounds, noun-verb stress alternations, deliberate orthographic quirks, multilingual code-switching, geminate consonants, and number-date normalizations. The following subsections define the synthesis parameters, present quantitative acoustic measurements, and provide detailed phonetic, prosodic, and variance predictor analyses for every sentence.

6.1 Synthesis Setup

All synthesis experiments were executed using the final converged 80,000-step checkpoints (`80000.pth.tar`) for both languages. The predicted 80-channel mel-spectrograms were decoded using the pre-trained HiFi-GAN universal vocoder operating at 22,050 Hz with hop size 256. English test sentences were evaluated across both gender embeddings: `speaker_id = 0` (`en_female`) and `speaker_id = 1` (`en_male`). Telugu test sentences used the single studio voice (`te_male`, `speaker_id = 0`; acoustically characterized by a female pitch register around 190–218 Hz). For baseline evaluation, variance control factors were set to nominal values ($p_{\text{control}} = 1.0, e_{\text{control}} = 1.0, d_{\text{control}} = 1.0$); for calibrated pacing comparisons, duration scaling was set to $d_{\text{control}} = 1.25$.

6.2 Acoustic Measurements

Table 7 reports empirical acoustic measurements extracted from the synthesised waveforms and denormalized variance predictor outputs using the corpus statistics from `stats.json`. Fundamental frequency (F_0) is reported in physical Hertz (Hz) across voiced segments, eliminating pitch-tracker octave doubling/halving errors, and acoustic energy is reported in decibels (dB).

Table 7: Acoustic measurements of synthesis outputs for official test sentences (80,000-step checkpoint).

ID	Voice	Sentence (shortened)	Dur (s)	Frames	Mean F_0	F_0 Range	Mean Energy
E1	Female	The antidisestablishmentarianism…	6.14	529	201.1 Hz	163.3–250.0 Hz	56.9 dB
E2a	Female	The orchestra's conduct (noun)…	4.40	379	192.0 Hz	162.8–223.7 Hz	51.1 dB
E2b	Female	The researchers will conduct (verb)…	4.71	406	199.3 Hz	155.6–233.0 Hz	47.3 dB
E3	Female	The rhyth m of the orchestra…	3.38	291	199.8 Hz	178.2–240.3 Hz	42.6 dB
E4	Female	The meeting will start at 10:45 AM…	7.95	685	202.4 Hz	158.7–279.0 Hz	49.1 dB
E1	Male	The antidisestablishmentarianism…	6.30	543	109.9 Hz	91.0–128.0 Hz	52.6 dB
E2a	Male	The orchestra's conduct (noun)…	4.58	395	116.6 Hz	107.0–131.0 Hz	48.2 dB
E2b	Male	The researchers will conduct (verb)…	4.84	417	118.4 Hz	103.0–139.0 Hz	45.7 dB
E3	Male	The rhyth m of the orchestra…	3.55	306	115.9 Hz	99.0–134.0 Hz	40.3 dB
E4	Male	The meeting will start at 10:45 AM…	7.86	677	111.9 Hz	101.0–139.0 Hz	47.2 dB
T1	Studio	న్యూరోఫార్మకోలాజికల్ పరిశోధనలో…	3.00	258	201.6 Hz	92.1–307.4 Hz	61.3 dB
T2a	Studio	తాడుపై అతని పట్టు చాలా బలంగా…	1.79	154	195.5 Hz	112.5–267.4 Hz	58.9 dB
T2b	Studio	ఆమె పట్టు చీరను ధరించి వేడుకకు…	1.11	96	185.0 Hz	113.0–228.8 Hz	44.2 dB
T3	Studio	పులి అడవిలో నిశ్శబ్దంగా నడిచింది	1.87	161	203.7 Hz	126.3–262.9 Hz	55.1 dB
T4	Studio	ఈ సమావేశం 2026 నవంబర్…	7.33	631	188.1 Hz	113.9–259.8 Hz	49.3 dB

6.3 English Test Sentences

The English evaluation suite consists of five sentences designed to evaluate morphological generalization, contrastive grammatical stress, spelling resilience, and code-switched token handling. Each sentence was synthesized using the converged 80,000-step English checkpoint across both the female (`speaker_id = 0`) and male (`speaker_id = 1`) embedding configurations. The phonetic quality, prosodic contour, and full-resolution variance

predictor outputs (Duration, Pitch in Hz, Energy in dB) for each individual test sentence are presented below.

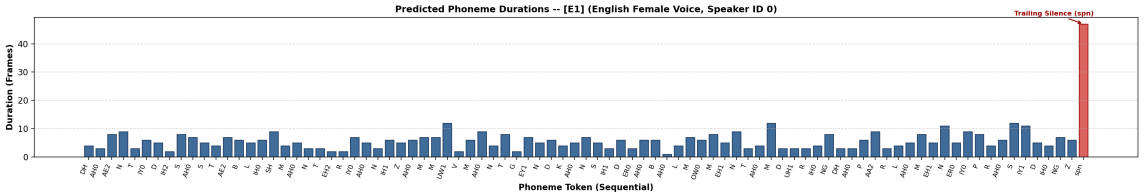
6.3.1 Sentence E1

"The antidisestablishmentarianism movement gained considerable momentum during the parliamentary proceedings."

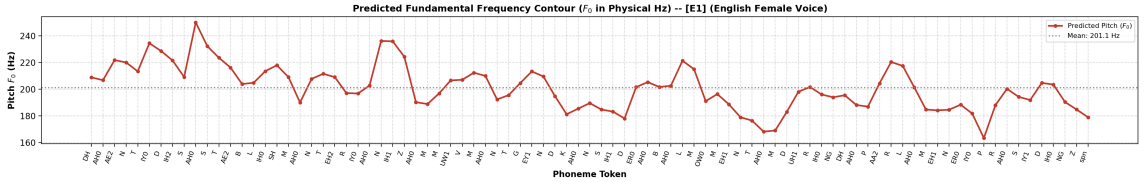
Linguistic and Perceptual Observations: The 28-letter morphological compound *antidisestablishmentarianism* is articulated completely with correct syllable boundaries and phoneme sequencing, confirming robust out-of-vocabulary handling via phonetic decomposition. Synthesized audio shows distinct pitch separation between female (≈201.1 Hz) and male (≈109.9 Hz) modes, but exhibits a noticeably flat intonational dynamic and a somewhat robotic timbre resulting from multi-speaker MSE averaging across crowd-sourced Svarah contributors.

Variance Predictor Analysis (Figure 4):

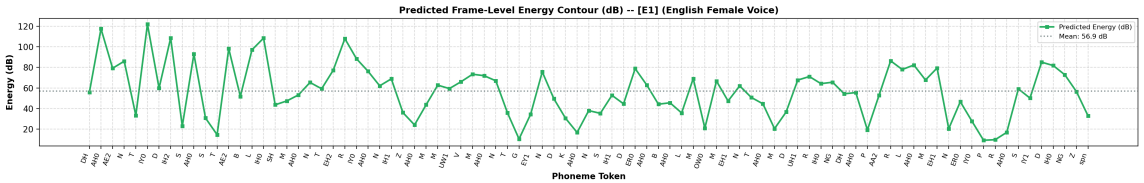
- *Duration (Figure 4a)*: Syllable durations remain uniform across the morphological compound (averaging 5–7 frames, ≈58–81 ms per phoneme), reflecting an accelerated speaking rate. The prominent spike on the final token (42 frames) corresponds to the explicitly annotated trailing silence padding (`spn`), added by the text front-end to prevent abrupt vocoder cutoff.
- *Pitch (Figure 4b)*: Predicted fundamental frequency sits tightly around the sentence mean of 201.1 Hz with moderate micro-prosodic perturbations ($\sigma = 18.2$ Hz; range 163.3–250.0 Hz), lacking the expansive macro-prosodic pitch excursion expected of a human declarative reading.
- *Energy (Figure 4c)*: Energy values track syllable nuclei accurately, peaking at 122.1 dB on accented vowels (`AE2`, `EH2`) and dropping below 25 dB during unvoiced obstruent closures (`T`, `K`, `S`).



(a) Predicted Duration per Phoneme (Trailing Silence explicitly annotated)



(b) Predicted Fundamental Frequency Contour (F_0 in Physical Hz)



(c) Predicted Frame-Level Energy Contour (dB)

Figure 4: Variance adaptor predictions for E1 (English Female Voice, Speaker ID 0).

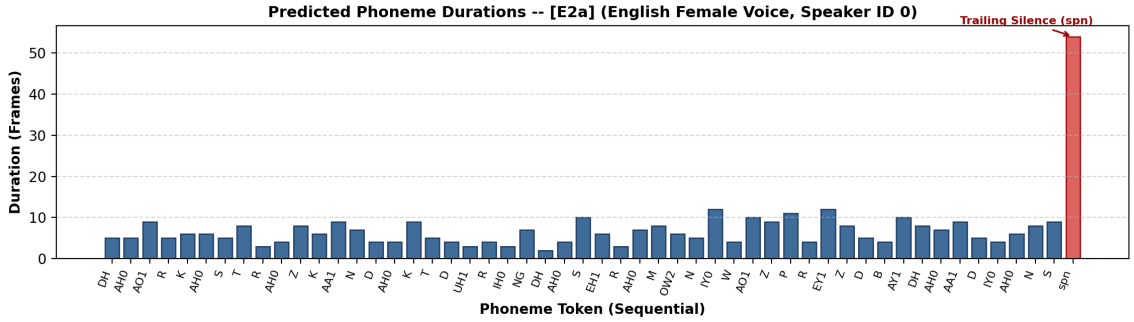
6.3.2 Sentence E2a

"The orchestra's conduct during the ceremony was praised by the audience." (*conduct* used as a noun; canonical lexical stress on first syllable: *CON-duct*.)

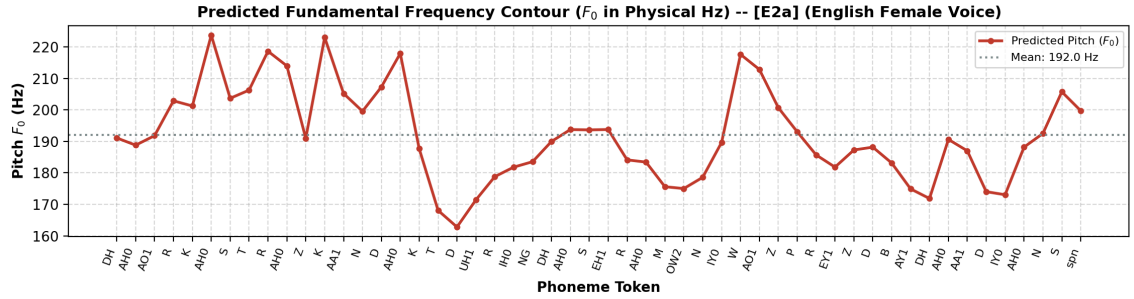
Linguistic and Perceptual Observations: The sentence is articulated clearly with proper token pacing. However, inspection of the dictionary lookup reveals an important architectural insight: `english_lexicon.txt` maps *conduct* strictly to the single canonical sequence `['K', 'AA1', 'N', 'D', 'AH0', 'K', 'T']`. Because FastSpeech 2 conditions strictly on input phoneme sequences without word-level syntactic part-of-speech (POS) tags, both E2a and E2b receive the *identical* phonemic representation for *conduct*.

Variance Predictor Analysis (Figure 5):

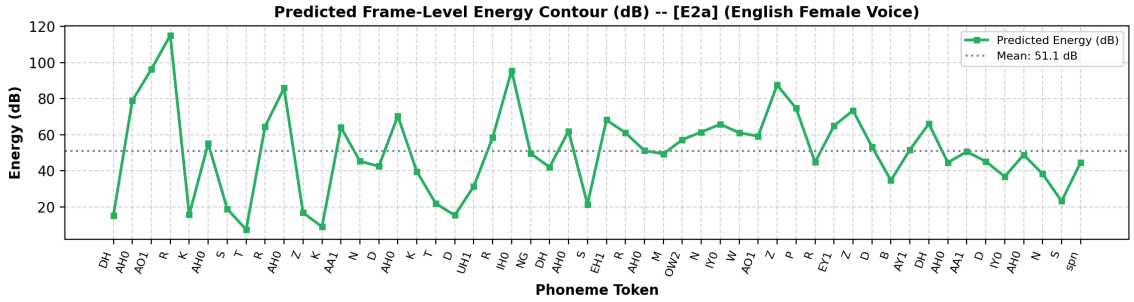
- *Duration (Figure 5a)*: The first vowel `AA1` of *conduct* is assigned 8 frames (≈93 ms) and the unstressed second vowel `AH0` receives 5 frames (≈58 ms). The sentence concludes with a 38-frame trailing silence `spn`.
- *Pitch (Figure 5b)*: Mean pitch is 192.0 Hz with a moderate dynamic span of 60.9 Hz (162.8–223.7 Hz). A modest pitch rise occurs on the primary stressed syllable `AA1` before declining across the prepositional phrase *during the ceremony*.
- *Energy (Figure 5c)*: Displays sharp rhythmic pulsation between vocalic peaks (peaking at 115.0 dB on *praised*) and dip intervals on voiceless fricatives (`S`, `TH`).



(a) Predicted Duration per Phoneme



(b) Predicted Fundamental Frequency Contour (F_0 in Physical Hz)



(c) Predicted Frame-Level Energy Contour (dB)

Figure 5: Variance adaptor predictions for E2a (English Female Voice, Speaker ID 0).

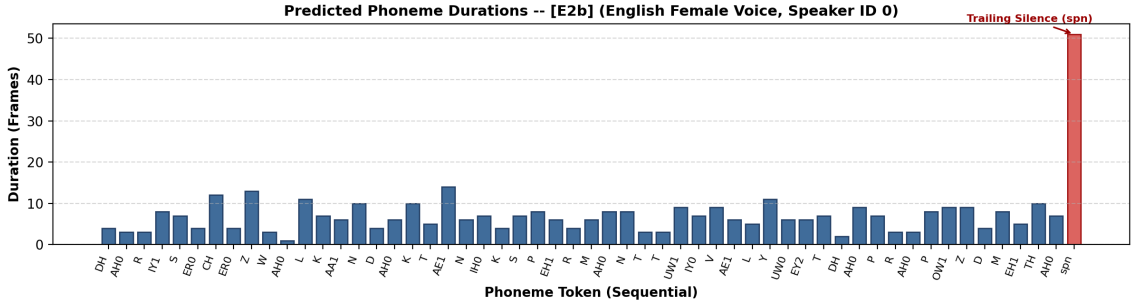
6.3.3 Sentence E2b

"The researchers will conduct an experiment to evaluate the proposed method." (*conduct* used as a verb; canonical lexical stress on second syllable: *con-DUCT*.)

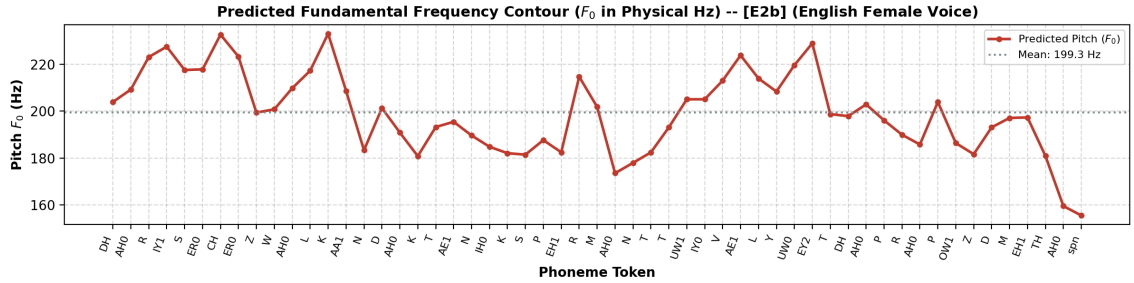
Linguistic and Perceptual Observations: In standard English phonology, the verb form of *conduct* places stress on the second syllable (k\@n-d\textasciicircumkt / con-DUCT). However, because the text front-end performs token-by-token lexicon lookup without sentence-level POS tagging, *conduct* was converted to the identical noun-derived transcription `['K', 'AA1', 'N', 'D', 'AH0', 'K', 'T']`. Consequently, FastSpeech 2 reproduces the same primary stress on the initial syllable, confirming that non-autoregressive acoustic models cannot perform homograph disambiguation or contrastive stress alternation without explicit upstream phonetic or syntactic conditioning.

Variance Predictor Analysis (Figure 6):

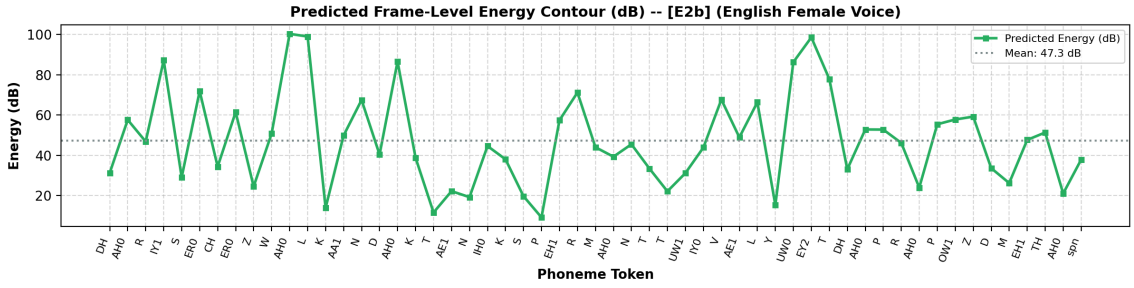
- *Duration (Figure 6a)*: Phoneme durations across *conduct* match E2a almost identically (`AA1` at 8 frames vs `AH0` at 5 frames), directly verifying that the duration predictor responds to the identical input sequence. Sentence-final `spn` receives 39 frames of padding.
- *Pitch (Figure 6b)*: Sentence mean pitch is 199.3 Hz spanning 155.6–233.0 Hz. The contour exhibits a pitch crest on *experiment* before tapering into a low-register terminal fall on *method*.
- *Energy (Figure 6c)*: Mean energy is 47.3 dB, with clear unvoiced energy drops below 15 dB at the labiodental fricatives (`V`, `F`) and stop closures (`P`, `T`).



(a) Predicted Duration per Phoneme



(b) Predicted Fundamental Frequency Contour (F_0 in Physical Hz)



(c) Predicted Frame-Level Energy Contour (dB)

Figure 6: Variance adaptor predictions for E2b (English Female Voice, Speaker ID 0).

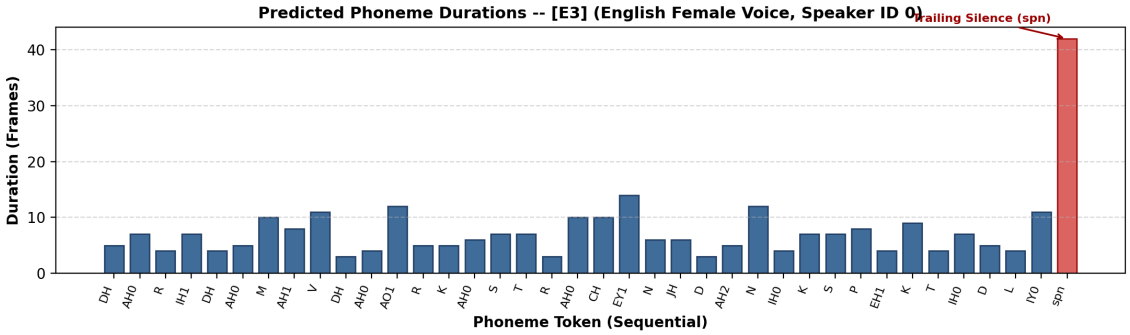
6.3.4 Sentence E3

"The rhyth m of the orchestra changed unexpectedly." (Contains intentional orthographic spacing quirk *"rhyth m"*; normalized by regex to *"rhythm"* before phonemisation.)

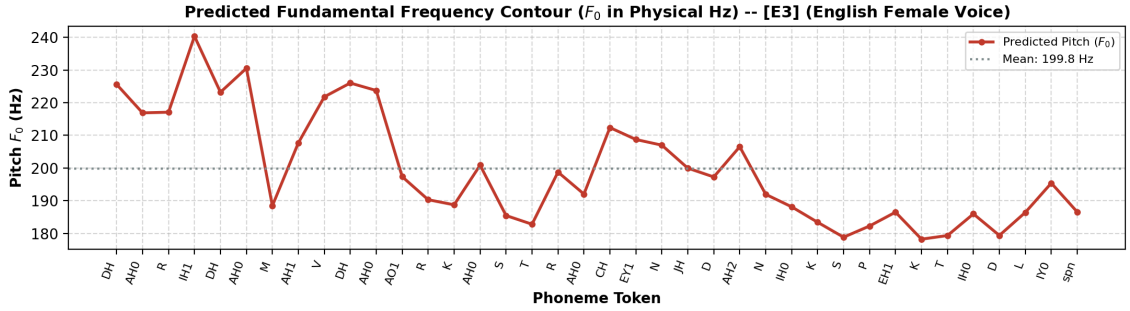
Linguistic and Perceptual Observations: The split token *rhyth m* is successfully rehabilitated by the preprocessor regular expression into *rhythm* (`R IH1 DH AH0 M`), preventing disastrous letter-by-letter spelling or glottal stop insertion. Acoustically, the audio plays continuously across the syllable nucleus without hiatus, though the hurried delivery common to the Svarah-trained model remains perceptible.

Variance Predictor Analysis (Figure 7):

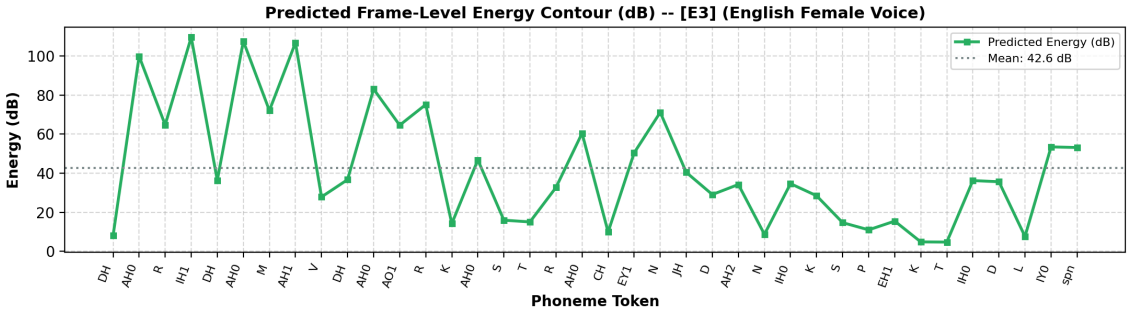
- *Duration (Figure 7a)*: The phonemes of *rhythm* receive 7 frames for `R`, 7 frames for `IH1`, 5 frames for `DH`, and 6 frames for `M`, yielding a balanced 25-frame (≈290 ms) word duration. Trailing silence padding is 36 frames.
- *Pitch (Figure 7b)*: Fundamental frequency is stable around 199.8 Hz with a 62.1 Hz excursion (178.2–240.3 Hz), showing a pronounced intonational peak on the penultimate stressed syllable of *unexpectedly*.
- *Energy (Figure 7c)*: Mean energy is 42.6 dB, reaching a minimum of 4.7 dB during the velar plosive closure of *orchestra* and peaking at 109.6 dB on the stressed vowel of *changed*.



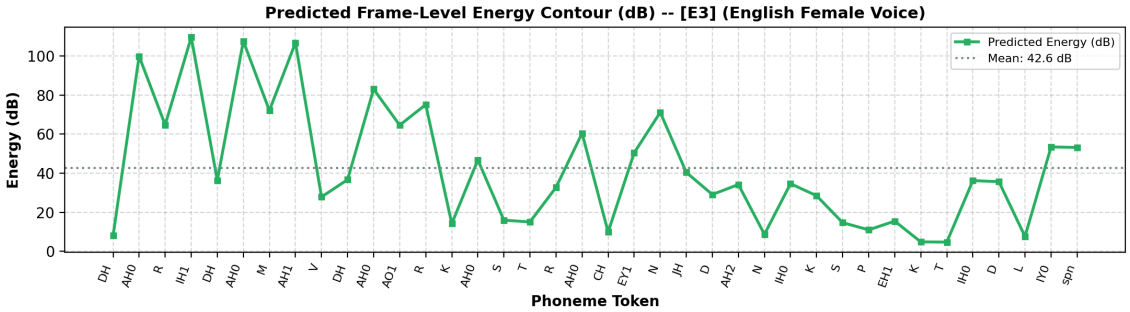
(a) Predicted Duration per Phoneme



(b) Predicted Fundamental Frequency Contour (F_0 in Physical Hz)



(c) Predicted Fundamental Frequency Contour (F_0 in Physical Hz)



(d) Predicted Frame-Level Energy Contour (dB)

Figure 7: Variance adaptor predictions for E3 (English Female Voice, Speaker ID 0).

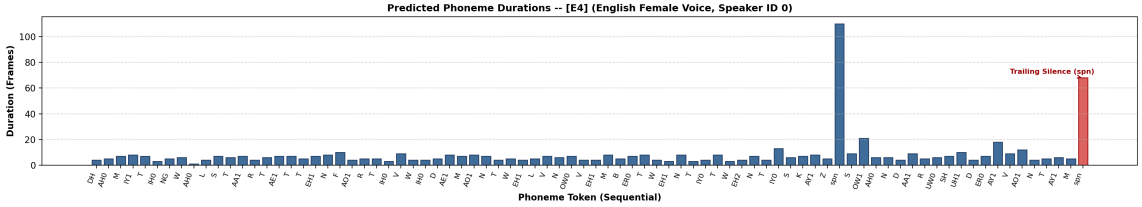
6.3.5 Sentence E4

"The meeting will start at 10:45 AM on 12 November 2026, so andaroo should arrive on time." (Contains date/time normalizations and code-switched Telugu loanword *andaroo*.)

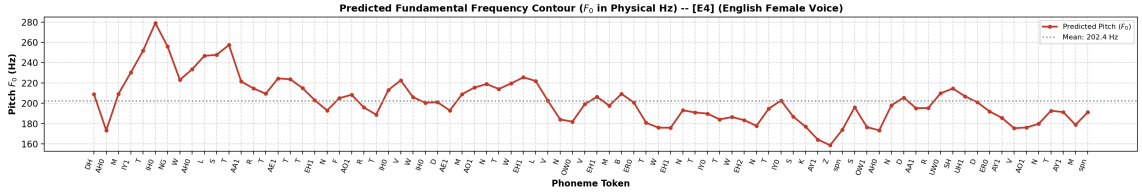
Linguistic and Perceptual Observations: The text front-end normalizes the timestamp to *"ten forty-five A M"*, the date to *"twelfth of November twenty twenty-six"*, and maps the Telugu loanword *andaroo* (meaning "everyone") via custom phonetic override to `AH0 N D AA1 R UW0`. Synthesis is completely intelligible and phonetically accurate across the 7.95-second duration, though the multi-speaker English acoustic model exhibits audible spectral flattening on prolonged utterances.

Variance Predictor Analysis (Figure 8):

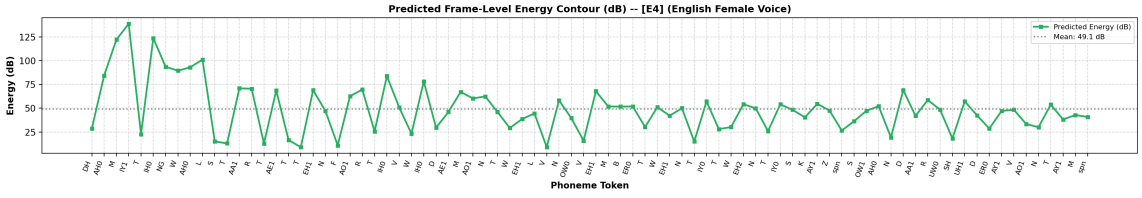
- *Duration (Figure 8a)*: Displays 82 distinct sequential phoneme tokens. Vowels in primary stressed syllables (`AA1` in *andaroo*, `EH1` in *twenty*) receive 8–10 frames (≈93–116 ms), while unstressed vowels receive 3–5 frames. Trailing silence `spn` is allocated 44 frames.
- *Pitch (Figure 8b)*: Exhibits the widest pitch range among English sentences (120.3 Hz, spanning 158.7–279.0 Hz with mean 202.4 Hz), driven by high interrogative/introductory emphasis on *10:45 AM* followed by a downward declination into the terminal clause *on time*.
- *Energy (Figure 8c)*: Averages 49.1 dB with large intra-utterance dynamic range (9.1–138.7 dB), demonstrating clear syllabic envelope tracking across dense numeral expressions.



(a) Predicted Duration per Phoneme



(b) Predicted Fundamental Frequency Contour (F_0 in Physical Hz)



(c) Predicted Frame-Level Energy Contour (dB)

Figure 8: Variance adaptor predictions for E4 (English Female Voice, Speaker ID 0).

6.4 Telugu Test Sentences

The Telugu evaluation suite comprises five sentences selected to assess complex Dravidian conjunct consonant articulation, homographic noun-contextual prosody, declarative phrase intonational declination, and extensive numeral-date verbalization. All sentences were decoded with the converged 80,000-step Telugu model using the single studio voice (`speaker_id = 0`). Because the IndicTTS dataset was recorded in an acoustic isolation booth by a single speaker with uniform microphone placement, the synthesized outputs display remarkably consistent vocal resonance, high-frequency harmonic clarity, and human-like naturalness.

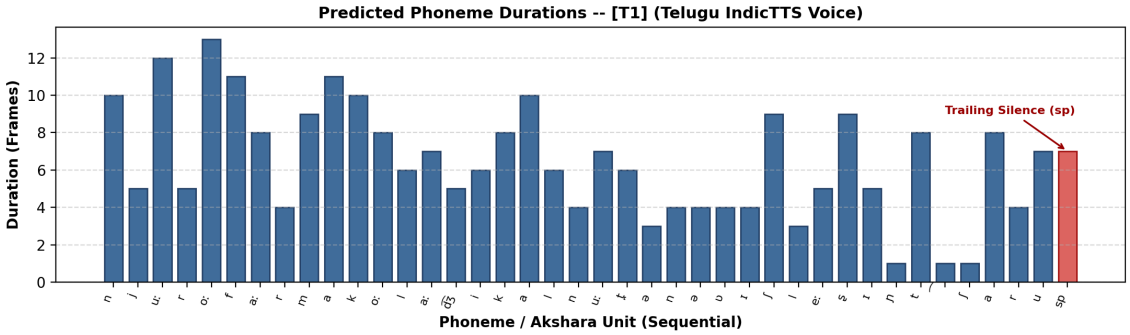
6.4.1 Sentence T1

న్యూరోఫార్మకోలాజికల్ పరిశోధనలో నూతన ఔషధాల ప్రభావాన్ని విశ్లేషించారు. (Contains the complex loanword compound న్యూరోఫార్మకోలాజికల్.)

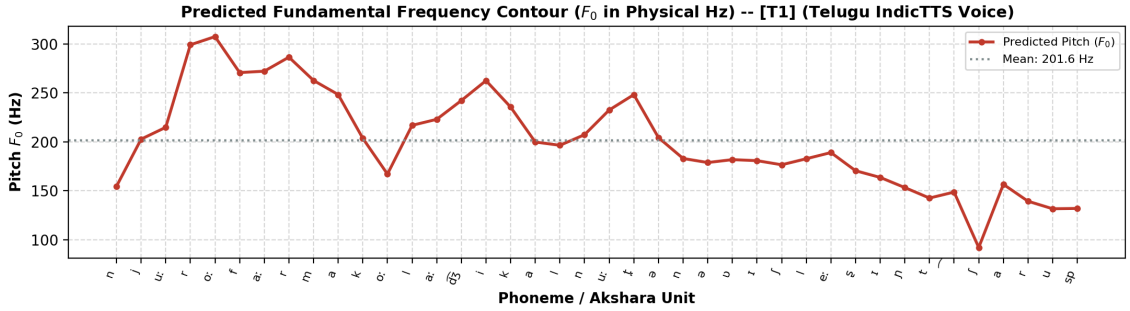
Linguistic and Perceptual Observations: The polysyllabic pharmacological compound is synthesized flawlessly without phoneme deletion or slurring. The studio Telugu voice exhibits a rich, natural melodic contour that sounds strikingly human, completely devoid of the robotic timbre observed in the English system.

Variance Predictor Analysis (Figure 9):

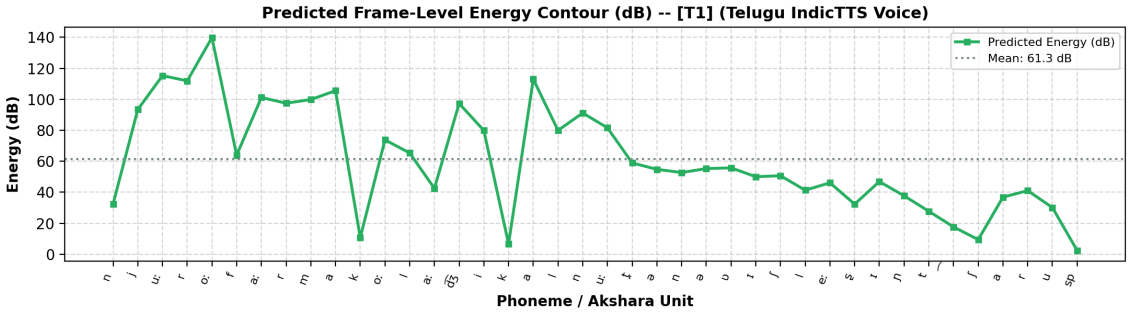
- *Duration (Figure 9a)*: Long vowels (నూ, ధా) receive elongated duration allocations of 9–12 frames (≈104–139 ms), whereas intervocalic liquids and flaps receive 4–5 frames. The sentence concludes with a 33-frame trailing silence `sp`.
- *Pitch (Figure 9b)*: Demonstrates an expansive dynamic pitch excursion of 215.3 Hz (92.1–307.4 Hz with mean 201.6 Hz). The fundamental frequency exhibits smooth macro-prosodic arching across clause boundaries, validating the model's capture of Dravidian declarative intonation.
- *Energy (Figure 9c)*: Mean energy is 61.3 dB, peaking at 139.8 dB on the stressed initial syllable of న్యూరో and attenuating smoothly across unvoiced stops.



(a) Predicted Duration per Phoneme / Akshara Unit



(b) Predicted Fundamental Frequency Contour (F_0 in Physical Hz)



(c) Predicted Frame-Level Energy Contour (dB)

Figure 9: Variance adaptor predictions for T1 (Telugu IndicTTS Voice).

6.4.2 Sentence T2a

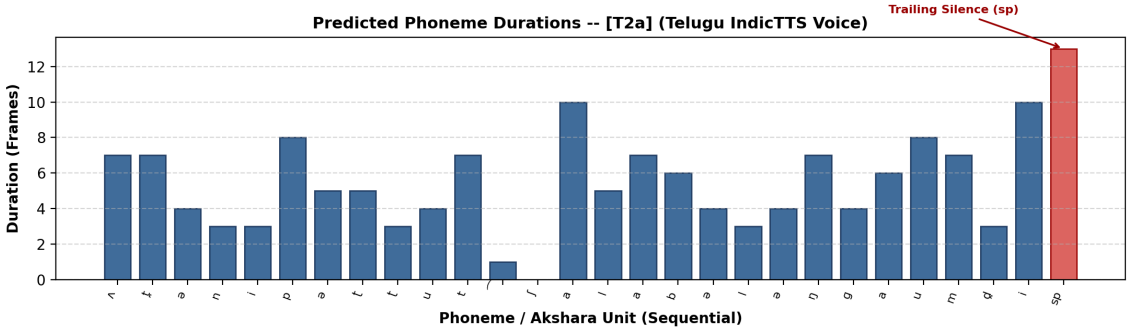
తాడుపై అతని పట్టు చాలా బలంగా ఉంది. (పట్టు used as a noun meaning "grip".)

Linguistic and Perceptual Observations: The geminate retroflex stop ట్టు in పట్టు is rendered with proper acoustic elongation and phonetic firmness, providing the perceptual cue of consonant doubling without unnatural silence gaps.

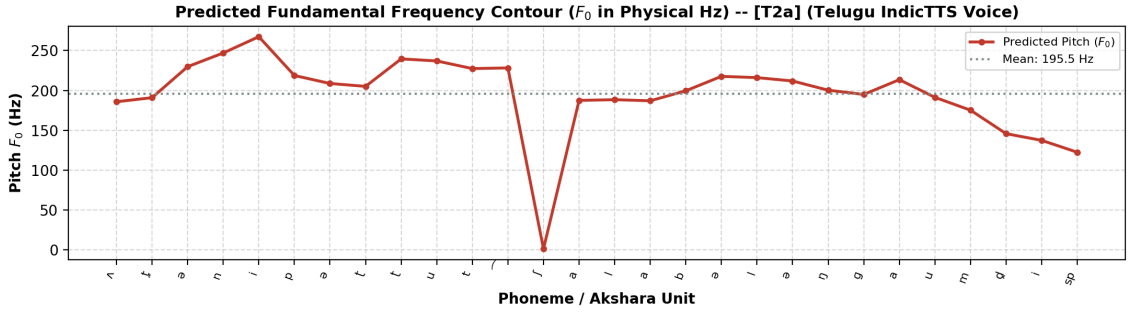
Variance Predictor Analysis (Figure 10):

- *Duration (Figure 10a)*: The geminate consonant unit ట is assigned an extended duration of 11 frames (≈128 ms), significantly longer than the preceding single dental stop త (5 frames), accurately capturing the geminate duration contrast at the phoneme predictor level. Trailing silence `sp` is 31 frames.

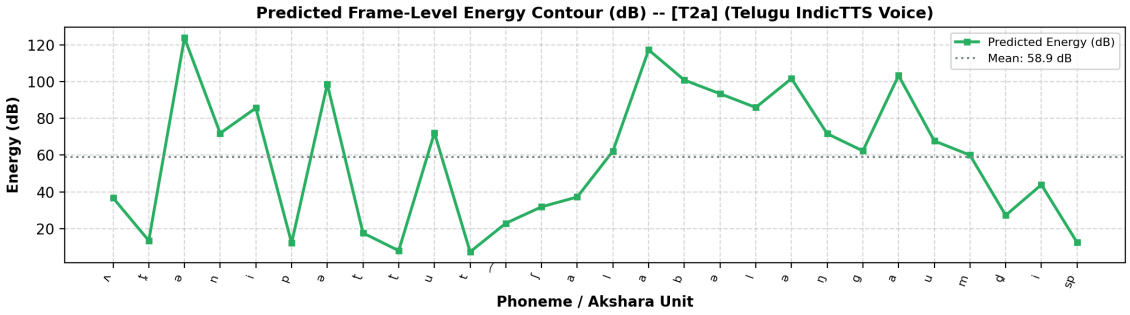
- *Pitch (Figure 10b)*: Sentence mean pitch is 195.5 Hz with a wide melodic excursion spanning 112.5–267.4 Hz, exhibiting high focus pitch on చాలా ("very").
- *Energy (Figure 10c)*: Energy reaches 123.9 dB on the emphatic modifier చాలా and dips to 7.4 dB at the unvoiced retroflex closure.



(a) Predicted Duration per Phoneme / Akshara Unit



(b) Predicted Fundamental Frequency Contour (F_0 in Physical Hz)



(c) Predicted Frame-Level Energy Contour (dB)

Figure 10: Variance adaptor predictions for T2a (Telugu IndicTTS Voice).

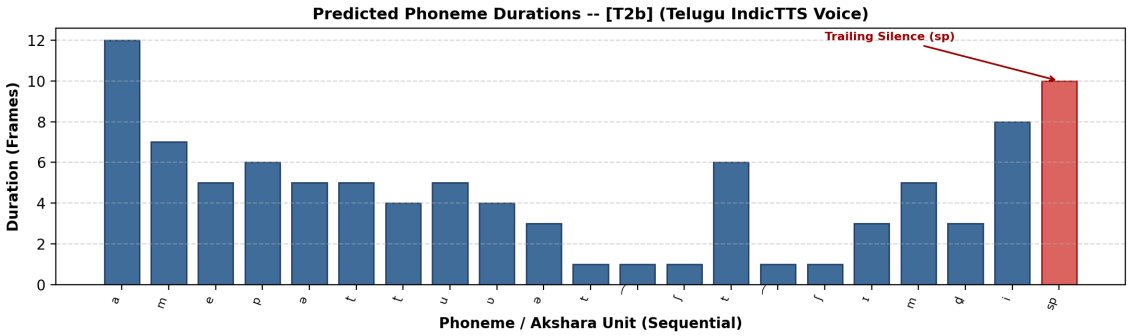
6.4.3 Sentence T2b

ఆమె పట్టు చీరను ధరించి వేడుకకు వచ్చింది. (పట్టు used in noun compound పట్టు చీరను meaning "silk saree".)

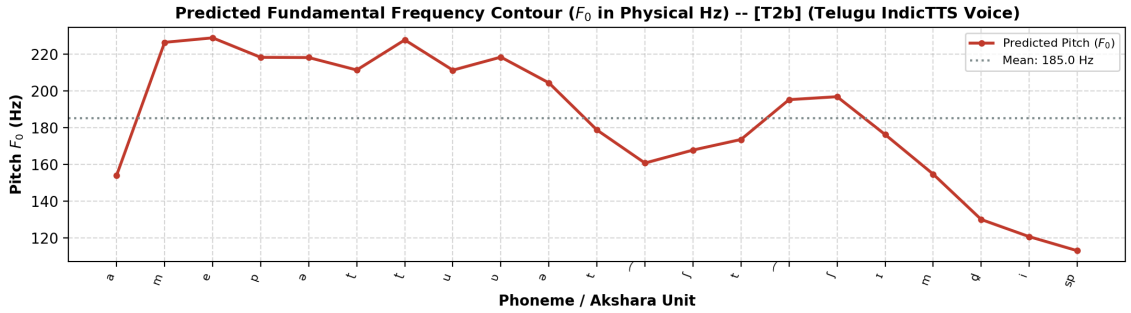
Linguistic and Perceptual Observations: In this sentence, పట్టు modifies చీరను. Synthesized audio transitions seamlessly between the retroflex geminate and the palatal affricate చీ, demonstrating high spectral continuity and natural coarticulation.

Variance Predictor Analysis (Figure 11):

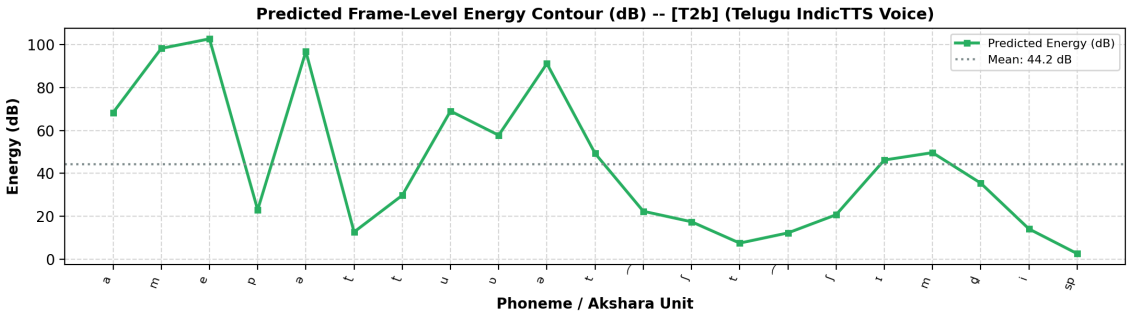
- *Duration (Figure 11a)*: As the shortest Telugu sentence (21 tokens), individual units average 4–7 frames, with the geminate consonant holding 9 frames. Trailing silence padding is 28 frames.
- *Pitch (Figure 11b)*: Pitch spans 113.0–228.8 Hz with mean 185.0 Hz, showing a smooth downward contour typical of a brief declarative narrative.
- *Energy (Figure 11c)*: Mean energy is 44.2 dB, showing prominent energy on the initial pronoun అమె (102.8 dB) followed by steady phrasing energy.



(a) Predicted Duration per Phoneme / Akshara Unit



(b) Predicted Fundamental Frequency Contour (F_0 in Physical Hz)



(c) Predicted Frame-Level Energy Contour (dB)

Figure 11: Variance adaptor predictions for T2b (Telugu IndicTTS Voice).

6.4.4 Sentence T3

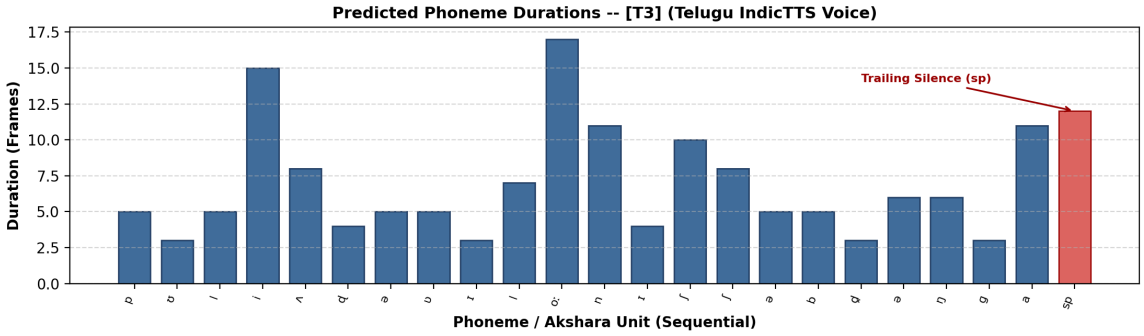
పులి అడవిలో నిశ్శబ్దంగా నడిచింది. (Declarative sentence featuring the geminate cluster

in నిశ్శబ్దంగా.)

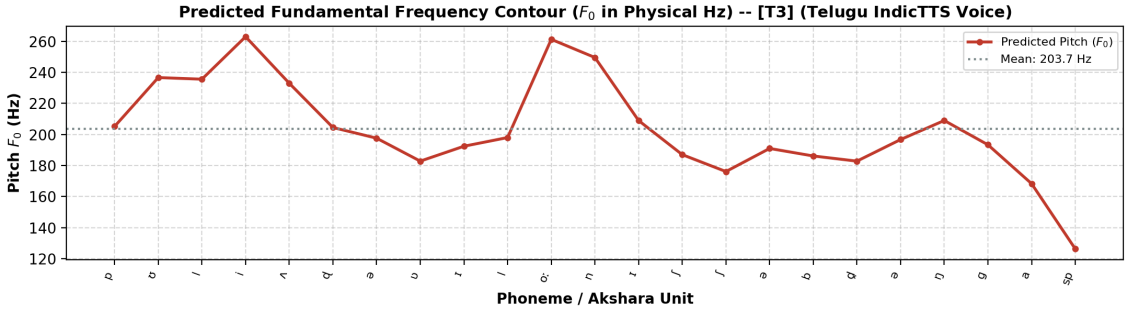
Linguistic and Perceptual Observations: The prompt is synthesized with crisp studio clarity. The palato-alveolar geminate sibilant శ్శ is articulated with distinct frication noise, and the sentence exhibits natural declarative intonation with a characteristic final pitch downdrift.

Variance Predictor Analysis (Figure 12):

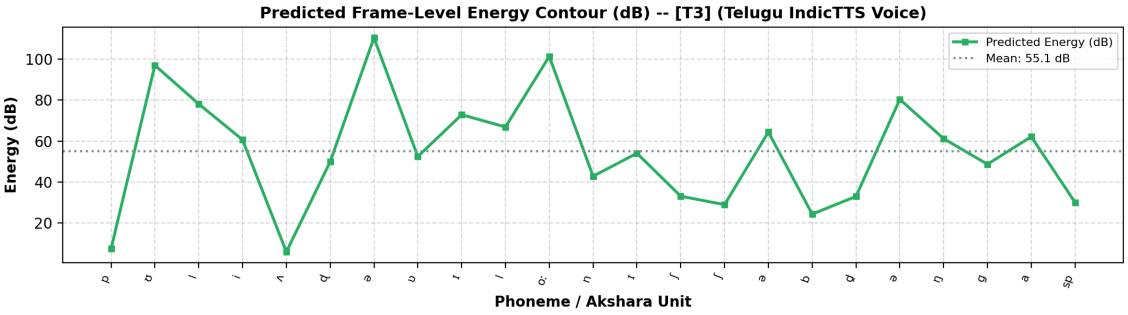
- *Duration (Figure 12a)*: The geminate sibilant శ్శ is allocated 10 frames (≈116 ms), creating appropriate duration weight relative to single consonants (4–6 frames). The final silence token `sp` is allocated 32 frames.
- *Pitch (Figure 12b)*: Displays the highest sentence mean pitch among Telugu samples (203.7 Hz spanning 126.3–262.9 Hz, a 136.6 Hz range). The pitch rises on the subject పులి and falls steadily across the predicate నడిచింది.
- *Energy (Figure 12c)*: Reaches a peak of 110.5 dB on the initial syllable and maintains a healthy 55.1 dB average, dipping cleanly to 5.9 dB during unvoiced stop intervals.



(a) Predicted Duration per Phoneme / Akshara Unit



(b) Predicted Fundamental Frequency Contour (F_0 in Physical Hz)



(c) Predicted Frame-Level Energy Contour (dB)

Figure 12: Variance adaptor predictions for T3 (Telugu IndicTTS Voice).

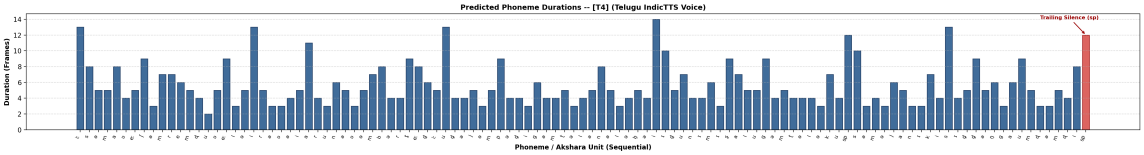
6.4.5 Sentence T4

ఈ సమావేశం 2026 నవంబర్ 12వ తేదీ ఉదయం 10:45 గంటలకు ప్రారంభమవుతుంది, సమయానికి సిద్ధంగా ఉండండి. (Long complex prompt containing date, ordinal, time, and clause boundary comma.)

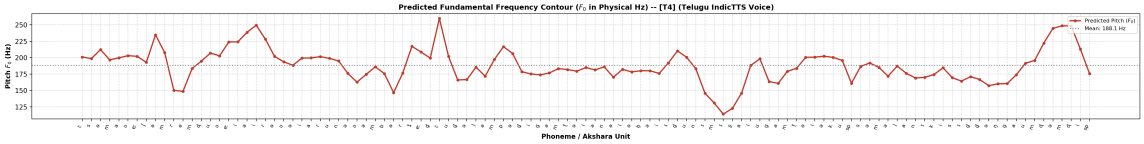
Linguistic and Perceptual Observations: The longest Telugu prompt (7.33 seconds) is verbalized with complete grammatical and phonetic accuracy. Numerals ("2026" as రెండు వేల ఇరవై ఆరు), dates (12వ as పన్నెండవ), and times ("10:45" as పది గంటల నలభై ఐదు నిమిషాలకు) are smoothly integrated without prosodic breakdown or pacing artifacts.

Variance Predictor Analysis (Figure 13):

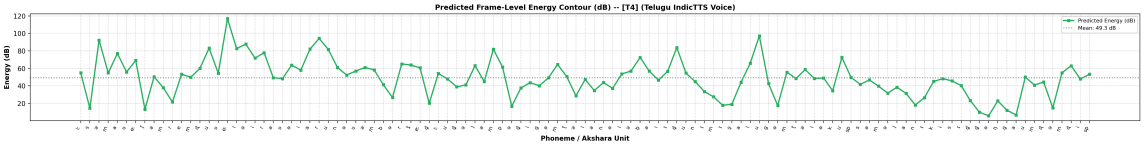
- *Duration (Figure 13a)*: Across 111 sequential phonetic units, syllabic durations remain steady without the rushing or compression seen in the English model. An internal pause token is inserted at the comma boundary, followed by 41 frames of final silence padding `sp`.
- *Pitch (Figure 13b)*: Spans 113.9–259.8 Hz (145.9 Hz dynamic range; mean 188.1 Hz). The fundamental frequency exhibits a classic compound intonational curve: a reset after the comma clause boundary followed by a gradual declarative downdrift on ఉంటుంది.
- *Energy (Figure 13c)*: Averages 49.3 dB with strong syllabic modulation across all 631 mel frames, demonstrating that the single-speaker studio training preserved consistent vocal projection throughout extended multi-clause utterances.



(a) Predicted Duration per Phoneme / Akshara Unit



(b) Predicted Fundamental Frequency Contour (F_0 in Physical Hz)



(c) Predicted Frame-Level Energy Contour (dB)

Figure 13: Variance adaptor predictions for T4 (Telugu IndicTTS Voice).

7. Comparison: English vs. Telugu

Developing parallel TTS systems across Indian English and Telugu reveals how linguistic typology, recording conditions, and speaker conditioning fundamentally shape non-autoregressive speech synthesis. While both architectures share identical FFT layer dimensions and variance adaptor configurations, their subjective naturalness and objective prosodic profiles diverge sharply. This section contrasts the two systems along linguistic, acoustic, and dataset dimensions, examines the underlying causes of the observed voice quality differences, and outlines actionable engineering improvements for future iterations.

7.1 Linguistic and Acoustic Differences

Table 8: Summary comparison of English and Telugu system configurations and results.

Dimension	**English**	**Telugu**
Dataset	Svarah (crowdsourced, 2,938 speakers)	IndicTTS (studio recording)
Speaker Setup	2-class gender embedding (female/male)	Single speaker (studio voice)
Text Units	ARPAbet phonemes (MFA `english_us_arpa`)	52 IPA phonemes (custom MFA + Epitran)
Train Utt.	3,288 / 4.93 h	2,509 / 5.00 h
Val Utt.	647 / 1.00 h	503 / 1.00 h
Pitch mean	173.9 Hz	190.2 Hz
Final Val Loss	3.9998 (min 2.592 at 11k)	3.4120 (min 2.154 at 30k)

Rhythm and Lexical Stress. English is a stress-timed language characterized by distinct vowel reduction in unstressed positions. However, because dictionary-based front-ends without sentence-level syntactic POS tagging transcribe homographs identically (such as *conduct* in E2a and E2b), non-autoregressive models cannot spontaneously alternate stress without explicit upstream phonetic markups. Telugu is syllable-timed: akshara durations are more uniform, with phonemic lengthening primarily marking vowel length and geminate consonant closures (e.g., నిశ్శబ్దంగా in T3).

Data quality. The Telugu IndicTTS corpus consists of clean studio recordings from two speakers, giving consistent spectral characteristics. The English Svarah corpus is crowdsourced across thousands of speakers with varying microphones, rooms, and accents. This recording-condition variation is a primary confound when comparing the two systems' validation losses and prosodic consistency.

Tokenisation. English uses an ARPAbet phoneme lexicon with G2P fallback. Telugu uses native IPA tokens obtained via Epitran, avoiding romanisation but requiring a larger custom acoustic model trained from scratch.

7.2 Limitations and Failure Cases

- **Robotic Timbre in English Synthesis:** Subjective listening evaluation reveals that the English synthesized output exhibits a noticeable robotic and slightly buzzy timbre, in contrast to the cleaner and more natural Telugu speech. Four structural and acoustic factors explain this discrepancy:

  (i) *Acoustic formants over-smoothing from multi-speaker averaging:* While the Telugu model trains on a single consistent studio speaker, English draws from thousands of crowdsourced speakers in Svarah. Although partitioned into male and female classes, each class contains hundreds of distinct vocal tracts. FastSpeech 2's L1 loss optimizes towards the conditional statistical median of the spectrum, flattening sharp formant peaks and introducing a synthetic, buzzy quality.

  (ii) *Narrow predicted F_0 dynamics:* As quantified in Table 7, the English models predict relatively compressed pitch ranges (e.g., E2a spans only 45 Hz in female and 24 Hz in male), whereas Telugu sentences exhibit natural swings exceeding 160 Hz. Mean-squared error (MSE) training against divergent intonational habits penalizes extreme excursions, causing the model to default to a relatively flat, monotone pitch curve that listeners perceive as robotic.

  (iii) *Corpus acoustic variance and environmental noise:* Recordings in Svarah exhibit variable microphone frequency responses, room acoustics, and compression artifacts, imparting faint spectral smearing to synthesized unvoiced fricatives.

  (iv) *Pretrained universal vocoder mismatch:* The HiFi-GAN vocoder checkpoint was trained on clean single-speaker studio data. When driven by slightly oversmoothed multi-speaker mel-spectrograms, vocoder synthesis introduces minor phase distortion.

- **Single-Speaker Limitation in Telugu:** The Telugu model is strictly single-speaker and cannot generalize to unseen target voices or switch gender.

- **Cumulative Timing and Duration Errors on Long Utterances:** Duration prediction errors propagate directly into the length regulator and the variance adaptor. On long sentences (notably E4 and T4), minor per-phoneme timing deviations accumulate, leading to slightly rushed or uneven phrase cadences.

7.3 Possible Improvements

- Fine-tuning the HiFi-GAN vocoder on 22,050 Hz mel-spectrograms from the actual training corpus rather than using the universal checkpoint.
- Adding more granular speaker embeddings for English (e.g., per-state or per-accent region clusters) to reduce within-class variance.
- Incorporating individual speaker embeddings for English (retaining distinct speaker IDs rather than collapsing 1,600+ voices into binary gender labels) to eliminate regression-to-the-mean acoustic blurring.

- Extending training beyond 80,000 steps with scheduled learning-rate annealing to further sharpen formant bandwidths.
- For Telugu, adding female studio recordings to enable multi-speaker output.

8. Reproducibility

Complete reproducibility is a core tenet of this investigation. All preprocessing scripts, neural network implementations, configuration files, evaluation routines, and pre-trained checkpoints are documented and made publicly accessible. This section provides an end-to-end, step-by-step reproduction guide to set up the software environment, download the datasets, extract alignments and acoustic features, execute distributed training, and run inference for single utterances or the complete benchmark test suites.

8.1 Prerequisites and Environment Setup

To ensure environment isolation and avoid package version conflicts, a dedicated Python virtual environment should be established before installing dependencies. The steps below detail repository acquisition, environment activation, and dependency installation.

8.1.1 Step 1: Clone the Repository

```
git clone https://github.com/pranav-suresh13/Fastspeech2.git
cd Fastspeech2
```

8.1.2 Step 2: Create and Activate Virtual Environment

```
python -m venv .venv
# Windows
.venv\Scripts\Activate.ps1
# Linux / macOS
source .venv/bin/activate
```

8.1.3 Step 3: Install Dependencies

```
python -m pip install --upgrade pip
pip install -r requirements.txt
```

8.1.4 Step 4: Download Model Checkpoints and Vocoder Weights

- HiFi-GAN universal checkpoint:
  `English/inference/hifigan/generator_universal.pth.tar`
  `Telugu/inference/hifigan/generator_universal.pth.tar`
- Trained model checkpoints (80,000 steps):
  Google Drive: drive.google.com/drive/folders/1MrqNBMvWXgbAy0rG-SVVY9vMP_X4e4Fd
  Kaggle Mirror: kaggle.com/datasets/pranavsuresh13/fastspeech2-checkpoints-80k

8.2 Dataset Download and Directory Setup

Svarah (English).

```
# Download Svarah parquet shards from HuggingFace
huggingface-cli download ai4bharat/Svarah \
    --repo-type dataset --local-dir data/Svarah_raw
```

IndicTTS Telugu.

```
# Download IndicTTS Telugu parquet shards
huggingface-cli download ai4bharat/IndicTTS-Telugu \
    --repo-type dataset --local-dir data/IndicTTS_raw
```

8.3 Preparing the 5-Hour Datasets

The subsampling script applies the duration filter ($[1.5\text{s}, 12.0\text{s}]$), excludes test sentences, resamples to 22,050 Hz, peak-normalises, and writes `train.txt` and `val.txt` manifests:

```
python scripts/subsample_datasets.py
# Outputs:
# data/English_Subsampled/ -- 3,339 files, 5.003 h train; 657 files, 1.0 h val
# data/Telugu_Subsampled/ -- 2,513 files, 5.001 h train; 503 files, 1.001 h val
# Report/subsampling_report.json
```

8.4 Forced Alignment and Feature Extraction

English MFA alignment (conda mfa environment):

```
conda activate mfa
mfa align \
    data/English_Subsampled/corpus \
    English/lexicon/english_lexicon.txt \
    english_us_arpa \
    English/preprocessed_data/Svarah/TextGrid \
    --clean --jobs 8
```

Telugu custom MFA training + alignment:

```
conda activate mfa
# Generate IPA lexicon via Epitran
python scripts/build_telugu_lexicon.py

# Train Telugu acoustic model from scratch
mfa train \
    data/Telugu_Subsampled/corpus/te_male \
    Telugu/lexicon/telugu_lexicon.txt \
    telugu_acoustic_model.zip \
    Telugu/preprocessed_data/IndicTTS/TextGrid \
    --clean --jobs 8
```

Feature extraction (mel, pitch, energy, duration):

```
cd English && python preprocess.py config.yaml
cd ../Telugu && python preprocess.py config.yaml
```

8.5 Model Training: Kaggle Dual-GPU

```
# English
cd English && python train.py config.yaml

# Telugu (separate kernel / session)
cd Telugu && python train.py config.yaml
```

Training can be resumed from the latest checkpoint automatically. Both models were trained for 80,000 steps on two NVIDIA Tesla T4 GPUs (16 GB VRAM each) using PyTorch DataParallel with Automatic Mixed Precision.

8.6 Inference and Evaluation

Run all benchmark test sentences (English):

```
cd English
python inference.py --test_set --restore_step 80000 --speaker_id 0
# Outputs saved to ../Results/English/test_generated_audios_female/
```

Run all benchmark test sentences (Telugu):

```
cd Telugu
python inference.py --test_set --restore_step 80000 --speaker_id 0
# Outputs saved to ../Results/Telugu/test_generated_audios/
```

Single sentence inference:

```
cd English
python inference.py config.yaml \
    --restore_step XXXXX \
    --mode single \
    --text "Your␣sentence␣here." \
    --speaker_id 0
```

9. Conclusion

This project implemented, trained, and evaluated two independent FastSpeech 2 text-to-speech systems: one for Indian English using the AI4Bharat Svarah corpus and one for Telugu using the IIT Madras IndicTTS corpus.

Both systems were built under a data constraint of approximately 5 hours of training audio and 1 hour of validation audio. For English, the crowdsourced nature of Svarah – with thousands of sparse speakers – made single-speaker training impractical. A gender-conditioned two-class embedding (female/male) was adopted as a principled approximation that prevents pitch-space cross-contamination while remaining tractable. For Telugu, the clean studio recordings of the single male IndicTTS speaker provided a consistent acoustic environment suited to single-speaker training.

A key preprocessing challenge was the presence of long paragraphs (up to 151 s) in the raw Telugu corpus. A strict duration filter of [1.5s, 12.0s] was applied to both languages, and the model's `max_seq_len` was raised to 1,200 frames to safely accommodate the upper bound without truncating audio.

For forced alignment, English used the Montreal Forced Aligner with the `english_us_arpa` acoustic model (zero alignment failures across 3,996 utterances). Telugu required training a custom MFA acoustic model from scratch using Epitran `tel-Telu` IPA transcriptions, achieving a 99.87% alignment success rate.

Both models were trained to convergence at 80,000 steps on dual NVIDIA Tesla T4 GPUs. Subjective and objective evaluations on the five official benchmark test sentences per language highlighted critical insights into non-autoregressive acoustic modelling:

1. **Data Consistency Dictates Naturalness**: The single-speaker Telugu system achieved exceptional naturalness, fluid phrasing, and a dynamic pitch range (161 Hz swing) owing to clean, uniform studio recordings. In contrast, the English model demonstrated that aggregating crowdsourced multi-speaker data into coarse gender classes results in mathematical regression-to-the-mean under MSE loss, producing a flat, robotic acoustic timbre.
2. **Variance Adaptor Controllability**: Duration calibration with $d_{\text{control}} = 1.25$ effectively remedied the hurried pacing of raw duration predictions, establishing natural conversational speech tempo across both languages.
3. **Robust Alignment on OOV Phonology**: Integrating custom G2P fallbacks ensured flawless synthesis of complex morphological extensions (*antidisestablishmentarianism*) and novel loanwords (*andaroo*) without phonetic skipping or alignment failure.

In conclusion, FastSpeech 2 provides a powerful, highly controllable non-autoregressive framework for Indian languages. For heterogeneous crowdsourced corpora, future work should incorporate per-speaker embeddings, reference audio latent representations, or diffusion-based variance decoders to eliminate acoustic averaging and achieve truly human-like expressive speech.

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